

Newsvendor Model

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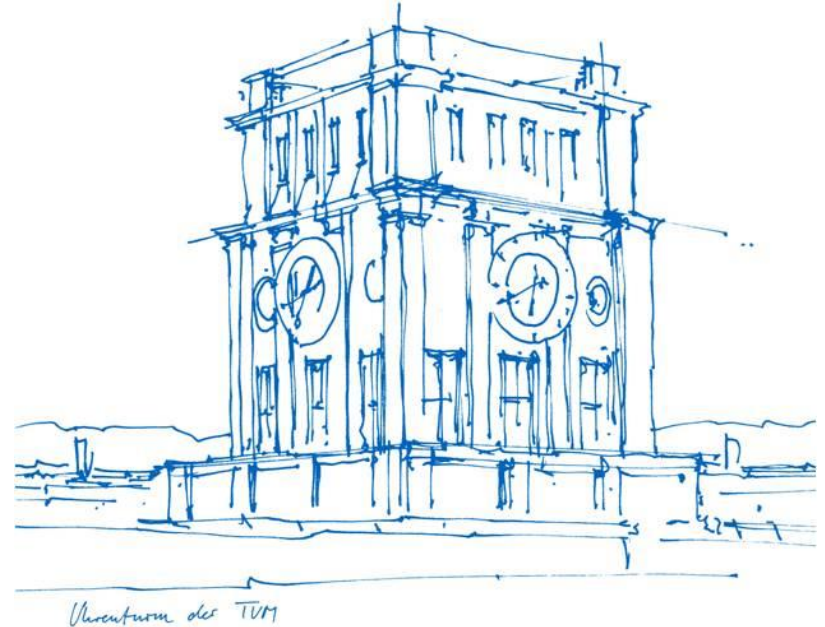
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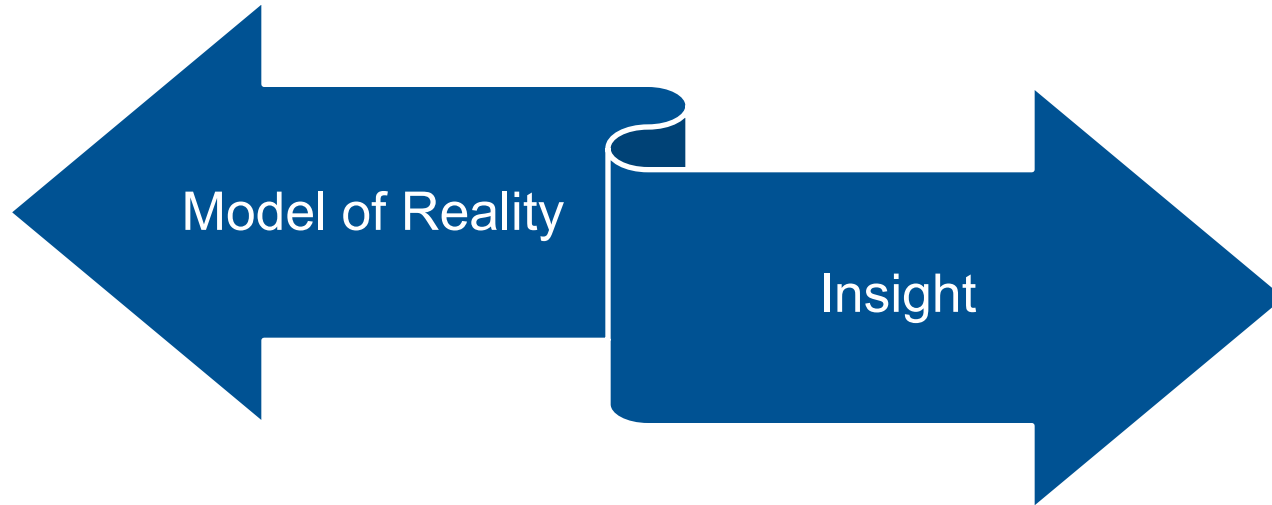
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Inventory Management with Uncertain Demand

Forecast, what's next?

Imagine you have forecasted a demand of 1,000 units and your forecast error is unbiased with a standard deviation of 50 units and it is normally distributed.

Further, you operate a make-to-stock business. That is, you need to decide how many units to order before you know the exact demand.

Unsold units cannot be salvaged.

Per unit sold, you make a profit of 10€. Per unit unsold, you incur a loss of 5€.

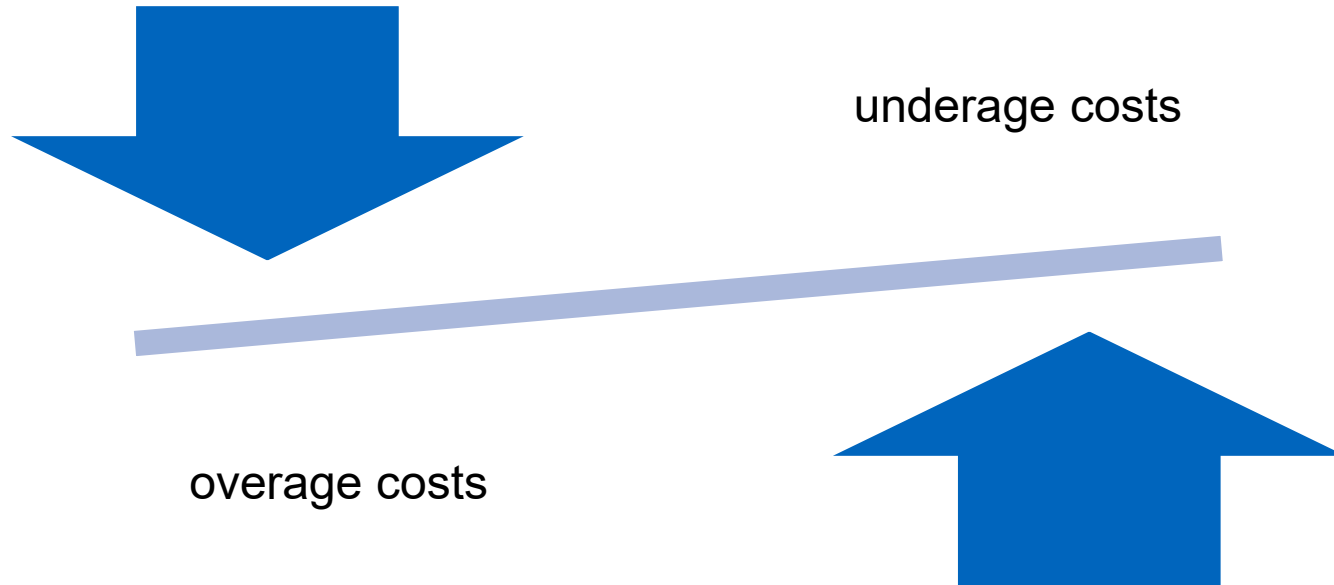
How many units should you order?

Newsvendor Model



How many newspapers should the newsvendor order from his distributor?

Fundamental Tradeoff in the Newsvendor Model



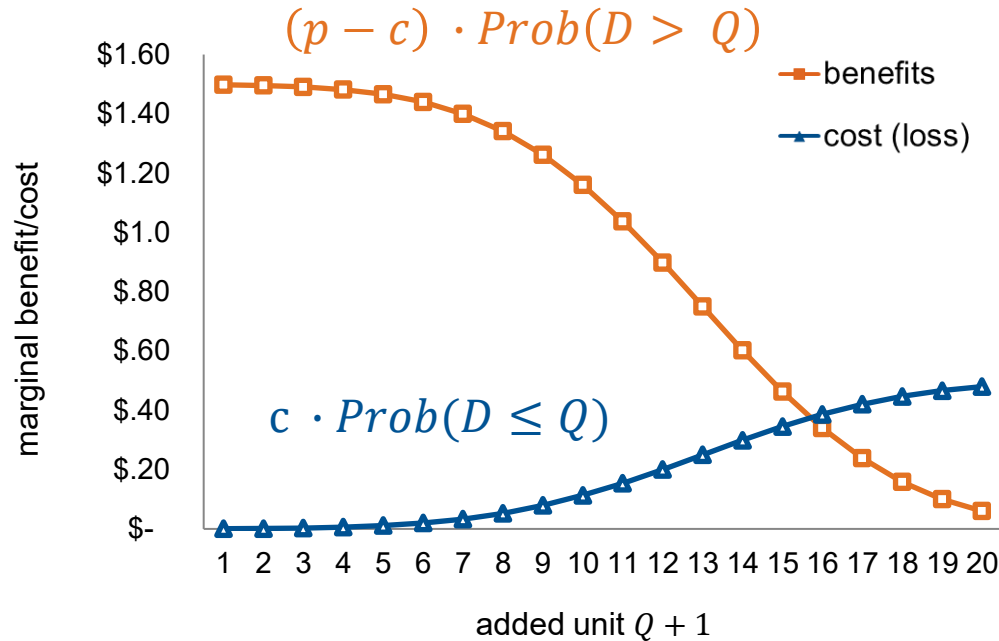
News vendor Solution

Optimal Order Quantity in the Newsvendor Model

Example with costs per newspaper $c = \$0.50$

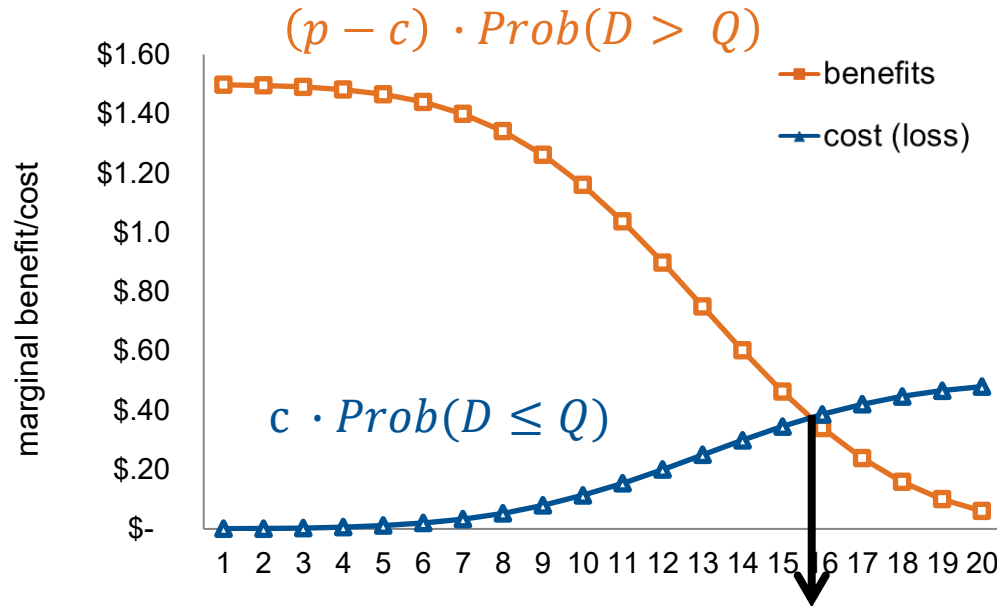
price per newspaper $p = \$2.00$

Optimal Order Quantity in the Newsvendor Model



Example with costs per newspaper $c = \$0.50$
 price per newspaper $p = \$2.00$

Optimal Order Quantity in the Newsvendor Model



Example with costs per newspaper $c = \$0.50$
price per newspaper $p = \$2.00$

$$(p - c) \cdot \text{Prob}(D > Q) = c \cdot \text{Prob}(D \leq Q)$$

Optimal Order Quantity in the Newsvendor Model

$$(p - c) \cdot \text{Prob}(D > Q) = c \cdot \text{Prob}(D \leq Q)$$

$$\Leftrightarrow (p - c) \cdot (1 - \text{Prob}(D \leq Q)) = c \cdot \text{Prob}(D \leq Q)$$

$$\Leftrightarrow p - c = c \cdot \text{Prob}(D \leq Q) + (p - c) \cdot \text{Prob}(D \leq Q)$$

$$\Leftrightarrow \frac{p - c}{p} = \text{Prob}(D \leq Q)$$

Optimal Order Quantity in the Newsvendor Model

The optimal quantity Q^* is the solution to $\frac{p-c}{p} = \text{Prob}(D \leq Q)$

The fraction $\frac{p-c}{p}$ is called the critical fractile. It represents the optimal service level (SL).

Let $F(Q) := \text{Prob}(D \leq Q)$, then $Q^* = F^{-1}\left(\frac{p-c}{p}\right)$

Key take-away: **The service level (SL) is the probability of meeting all demand.**

A (Weak) Generalization to Underage and Overage Costs

$$\begin{array}{rcl} & \text{underage costs} = & c_u := p - c \\ + & \text{overage costs} = & c_o := c \\ \hline = & & c_u + c_o = p \end{array}$$

$$\Rightarrow \text{Prob}(D \leq Q) = \frac{p - c}{p} = \frac{c_u}{c_u + c_o}$$

“Formal Proof”

Redo the step-by-step analysis as before but with underage and overage costs.

Formal proof: calculate expected values and use Leibniz integral rule for solving them (quite technical).

Applying the Newsvendor Model

Newsvendor Model: Interpreting the Solution



Service level = 50%

Newsvendor Model: Interpreting the Solution



Service level $< 50\%$

News vendor Model: Interpreting the Solution



Service level $> 50\%$

Newsvendor Model: a Simple Example

Demand: Uniform and continuous between 2,000 and 5,000. That is for any quantity between 2000 and 5000,

$$Prob(D \leq Q) = \frac{Q - 2000}{3000}$$

Costs per news paper: $c = \$0.50$

Price per news paper: $p = \$1.50$

What is the optimal service level and how many items should be ordered?

Newsvendor Model: a Simple Example

Solution

$$\text{Service level} = \frac{p - c}{p} = \frac{1.50 - 0.50}{1.50} = \frac{1}{1.5} = \frac{2}{3}$$

$$\text{Prob}(D \leq Q) = \frac{Q - 2000}{3000} = \frac{2}{3}$$

$$\Rightarrow Q^* = 4000$$

Forecast, what's next? – The initial example

Imagine you have forecasted a demand of 1,000 units and your forecast error is unbiased with a standard deviation of 50 units and it is normally distributed.

Per unit sold, you make a profit of 10€. Per unit unsold, you incur a loss of 5€ $\rightarrow SL = \frac{10}{10+5} = \frac{2}{3}$.

How many units should you order? $\rightarrow F^{-1}\left(\frac{2}{3}; \mu = 1000, \sigma = 50\right) = 1021.54$ which we round down.
Hence 1,021 units should be ordered.